

Sway Together, Stay Together: Visualizing Spiritual Support Networks Through the *SoulGarden* Prototype

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ABSTRACT

Spiritual support is a fundamental yet understudied dimension of support exchange in Online Health Communities. Recent work suggested that visualization of spiritual support networks offers a promising design mechanism for helping patients and caregivers navigate life-threatening physical or mental illnesses. Therefore, through user-centered and community-situated inquiry, we designed and developed *SoulGarden*—an interface that helps users visualize their spiritual support network in order to reduce loneliness and increase a sense of community and belonging. Initial feedback from user tests suggests that our nature-based theme provides users with feelings of calm and comfort, and that seeing the faces and prayers of supporters adds a meaningful sensory component to online support exchange. Our evaluation also suggests three valuable design considerations for future support network visualizations.

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1 INTRODUCTION

Building on the rich history of studying social support in Social Computing and Human-Computer Interaction (e.g., [2, 5, 12, 13, 19, 21]), Smith et al. shine a new theoretical lens on support exchange in their 2021 CSCW paper [16], defining *spiritual support* as a critical dimension that underlies and deeply influences people's experiences in online health communities. In particular, CaringBridge (www.caringbridge.org) is an online journaling platform that facilitates virtual support communities for patients and caregivers [8, 11, 17]. Participatory design focus groups with CaringBridge stakeholders suggested four design opportunities to help users feel more spiritually connected with their support networks,

and to find the inner strength to cope with life-threatening or terminal illnesses [16]. In this work, we explore the design implication that: **designers should create *support network visualizations* that empower support recipients by showing them the continual and interactive presence of their supporters and communities.**

Addressing this challenge with a user-centered and community-situated approach, we first conducted a focus group at a local Christian church. We did not want to constrain our efforts within the existing CaringBridge interface. Instead, we asked participants to conceptualize ways in which we might represent a user's spiritual support network through any online interface. Next, we analyzed the data, synthesized design implications, and consulted with a visualization expert to gain professional recommendations for our design. We then developed *SoulGarden*—a prototypical web implementation that lets users visualize their spiritual support network as an interactive garden of colorful swaying flowers, each with the image of a supporter's face at its center. In this work, we describe our user research, design and implementation, and user testing of our prototype. Participants' feedback on key elements of our design and its primary features provides valuable insights for future research to help patients and caregivers experience a loving sense of spiritual support through the presence of their community in an online space.

2 USER RESEARCH

We conducted an in-person focus group during January 2020 with 9 members of a local Christian church after a Sunday service. We did not formally collect demographic information, however most participants were middle-aged and white; 7 presented as female. We facilitated three activities:

- (1) **Free sketching:** We presented participants with design questions and asked them to sketch their own intuitive ideas for a spiritual support network visualization. (See Figure 1 for example sketches.)
- (2) **Reflection on designs from prior literature:** We showed participants examples of the following three visualization concepts and engaged them in a discussion to gather feedback:
 - **Social closeness [20]:** The user is shown within concentric rings of supporters categorized by closeness (e.g. family, friends, acquaintances). Icons represent the support type each supporter provides (e.g. empathy, advice).
 - **Geography [16]:** The user is shown as the focal point on a map of the United States with supporters mapped around

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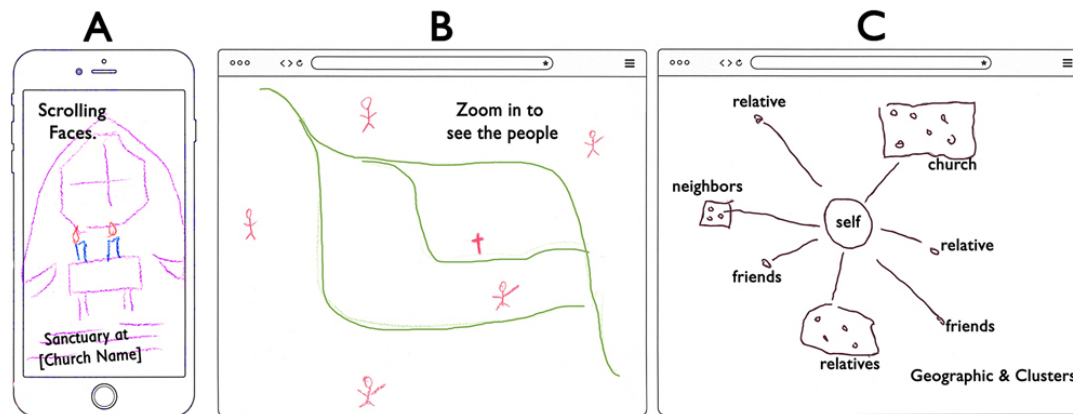


Figure 1: Participant sketches from a focus group at a local church. Sketches have been digitally enhanced; original handwriting replaced with typeset text for improved legibility.

them based on geographic location. When a supporter is online, their pin glows to indicate “presence”.

- **Face focused [15]** : A scrolling list view of supporters’ faces.
- (3) **Ranking & discussing favorites**: Participants completed a worksheet to rank their top 4 favorite visualizations from (1) and (2). They wrote down things they liked/disliked about each design, then discussed as a group.

3 FOCUS GROUP FINDINGS AND IMPLICATIONS FOR DESIGN

We collected 7 completed sketches and voting worksheets (since some participants worked together), transcribed audio recorded during the session, and inductively analyzed the data to capture emergent themes. The sketches showed a diverse array of possible interfaces, all of which prominently featured church and other community members arranged in meaningful places, larger geographic areas, or as conceptual clusters, and some of which represented actions such as prayer, holding hands, or gathering community members around the patient for hugs/physical touch. Figure 1 depicts three representative sketches. For example, one participant drew a lit candle with the image of the church’s sanctuary in the background. She explained that her visualization would show a list of scrolling faces atop that background, since seeing both her sanctuary and loved ones brings her joy and comfort (Figure 1(A)). Another participant drew a map of the United States; her church was marked with a cross symbol and her support network was represented by stick figures across the country. She wanted to visualize how broad her network was, with the ability to zoom in on specific areas to see supporters’ faces (Figure 1(B)). Another sketch showed the user surrounded by clusters of different communities that provide them with support (church, relatives, friends, neighbors, etc.), also with the ability to zoom in (Figure 1(C)).

No clear “winner” emerged from the ranking activity, and all visualizations received similar average scores. However, common themes emerged as participants discussed their rankings. One participant said, “*You move between needing support, giving support, or both at once.*” To them, it was important to evaluate designs both

from the perspective of a patient or caregiver receiving support and a supporter providing their support. Reflecting on the social closeness visualization, another participant said, “*I don’t know how it helps to categorize supporters. Sometimes, people not so close to you become more supportive than the people you’d expect.*” Participants found concentric circles to be overwhelming and felt that it was unnecessary to categorize supporters by social closeness. Reviewing geographical visualizations, they appreciated being able to see their network in a broader light to reiterate the strength of the support received from all over the country. However, one participant mentioned an important shortcoming—“*One thing that concerns me is that when you look online and nobody is lit up, it doesn’t mean that nobody is caring about you at the moment.*” Participants expressed concerns about feeling disappointed if none of the supporter pins were currently glowing, i.e., no one was online when they viewed their visualization. Finally, many participants emphasized how crucial it was for them to see the actual faces of their supporters to feel that they were “*real.*”

In conclusion, participants desired a simple visualization with supporters’ faces meaningfully arranged to provide a sense of physical connectedness, community, and belonging. Our analysis offers these key implications for design:

- The visualization should provide a sense of comfort, community and belonging.
- The visualization theme should be a meaningful and uplifting physical space or metaphor.
- Faces are critical in support networks to be able to form a connection with the supporters.
- Supporters should be identified based on both their group or community and as individuals.
- Users must continually feel supported in times of need – “*presence in absence.*”

4 EXPERT CONSULTATION TO SELECT A DESIGN APPROACH

We discussed our implications with Prof. Daniel F. Keefe, who specializes in interactive and artistic visualizations. His evaluation



Figure 2: Prototypical web implementation of *SoulGarden*. From left to right, clicking icons in upper right corner results in: addition of a new supporter to the garden; addition of a new “prayer” petal to an existing flower; change in all flower colors (hues of purple or orange); simulate passage of time (petals fade from a darker to lighter hue). A video demo is available at <https://youtu.be/fiotU3sHbF8>.

suggested that our design context/goals called for a solution that was aesthetically pleasing and uplifting, focused more on an immersive and ambient visual experience than typical task-based user interfaces. Inspired by this expert input and the work of Giorgia Lupi [9, 10], which focused on designing interactive visualizations that were “unique, contextual and intimate” and encouraged people to build digital connections, we decided to represent our visualization as a garden of flowers. Prior research also provides evidence that prayer support (“prayers, spiritual blessings, positive karma, good juju, warm thoughts”) is of the highest importance to many online health community users [17]. Working with a community of faith, our work accentuates the vital role that prayer plays in users’ well-being and online experiences. We chose to represent prayers or warm thoughts as flower petals, since nature can transcend specific religious or spiritual boundaries. Because participants wanted to include people from different parts of their life, this choice makes *SoulGarden* accessible to people both within and beyond the church. Furthermore, flowers have traditionally been used to convey well-wishes and support during health crises, and have many attributes (e.g. color, height, size, number of petals, etc.) that can convey valuable emotional information in a visualization [18].

5 IMPLEMENTATION OF SOULGARDEN

We implemented a web-based interface using HTML, CSS, JavaScript and p5.js (Figure 2). We picked an image with trees and a shining sun as the background [1]; higher fidelity prototypes could allow users to upload a background image of personal significance. Each supporter was represented by a swaying flower with their face in the center. Rather than a static flower, we added a swaying motion to provide a more engaging and immersive visual experience. We experimented and picked reasonable values for flower attributes to convey supporter information. Number of petals represented the number of times a supporter prayed; this began at 3 and was capped at 15 to maintain flower shape. Six tints of the same color (purple or orange [3]) were used to show how *recently* supporters

had prayed; the darkest shade represented within the last hour, and the lightest within the last year. Flower height represented the length of the supporter-user relationship. Stems of supporters from the same community (e.g. church, neighbors, etc.) were rooted near one another to suggest relational clusters. Users could hover over flowers to view more supporter details.

We envision that the default visualization of swaying flowers could be ambiently present (e.g. on a hospital room TV or automatic screensaver) or accessed online at any time. Whenever a key action occurred, the garden visualization updated to catch the user’s attention. For instance, when a new supporter joined the garden, a new flower appeared and flashed yellow for 15 seconds. When an established supporter prayed, a new petal was added to their existing flower, which flashed yellow for 15 seconds to show current presence. Given our preliminary user research, we did not want users to feel alone if no flowers were currently flashing, therefore we chose to fade individual flower colors gradually with the passage of time, and to reset the hue to full vibrancy whenever a new prayer arrived. This fading was intended to allow users to view and compare which groups of people were currently or recently online and praying.

6 EVALUATION AND FUTURE WORK

In April 2020, we conducted three user test sessions over Zoom (due to the COVID-19 pandemic) to solicit feedback from 5 users with experience with health crises (e.g. cancer). We performed live demos of events in the *SoulGarden* interface: when a new supporter joins the garden, existing supporters pray, or time passes. **Overall, participants were excited about the possibility to use the visualization themselves, or recommend it to family and friends facing health crises.** They found the garden theme soothing and spirit-elevating, describing a sense of happiness and vigor as they looked at the swaying flowers. They liked the use of the color yellow to draw attention to changes in the garden since they associated

it with connection to a higher power, or a blessing to hasten healing. They were also enthusiastic to creatively enrich the nature metaphor. For instance, a slow-blooming seed could signify the arrival of a new supporter, or flying birds might also signify the arrival of a new prayer. To ensure that the visualization achieves its basic purpose while remaining sustainable [4], future work should carefully balance which features to implement and which to omit in order to create an aesthetically pleasing and uncluttered experience.

Participants' concerns also highlighted three valuable design considerations. One participant worried about unwanted supporters or "weeds." **(1) She suggested the option to approve/disapprove requests to join, highlighting the vital importance of privacy controls in health communities, as in [16, 17].** Another participant was concerned that sending prayers could become like a task on her to-do list, or a time-saving alternative to *real* care. The concept of "*selfish shortcuts*" that benefit support providers more than recipients has been recognized as a wicked design problem [6]. Rather than a simple button press, this participant wanted to meaningfully personalize prayer petals with a message, image, or audio/video recording. **(2) This suggestion highlights the importance of quick yet high-quality connections through enriched media formats.** Finally, in our demo, all flowers faded at the same rate (without new incoming prayers), which is equivalent to a scenario in which all supporters stopped sending prayers at once. Participants thus felt that the fading feature could cause feelings of sadness and disappointment for patients/caregivers, and might induce guilt in supporters for not logging in enough. We view this feedback as a limitation of our demo, which did not allow users to view and compare how recently different individuals or groups had prayed. Nonetheless, participants greatly valued the addition of petals and the flashing feature when new prayers arrived. **(3) These observations suggest that representations of user actions (e.g. prayers) should emphasize simultaneous presence online, and should also exercise caution in designing features that visually diminish past actions.** Future research should evaluate mechanisms for motivating supporters without feeling "guilted" into it.

Designers should engage in critical reflection [14] as they expand on the *SoulGarden* concept to create more personalized and interactive visualizations. We worked with just one community of faith; future work should engage with user groups of diverse ages, genders, races, and beliefs. Future work must also evaluate mechanisms for visual scalability as the supporter base grows, such as: imposing an upper limit on the number of supporters; periodically cycling through supporters on screen; or enabling different screens for different communities. Furthermore, deployment of a higher fidelity prototype would yield even deeper insights about users' interactions and reflections (e.g. behavioral logs, Likert ratings on specific behaviors or emotions) [14]. Beyond health crises, spiritual support network visualizations may also be generally useful for enhancing social connectedness—e.g., *SoulGarden* might help people feel continuously supported when physical interaction is restricted due to a pandemic, geographic separation, or immobility. Whereas prior work has presented visualizations based on conventional task-based user interfaces or graph theory models, we hope that our proof-of-concept interface encourages researchers

to seek new creative and contextual solutions that honor the vital importance of spirituality in online interactions.

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